

Name _____

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Use the diagram given to answer questions 1 & 2.

1. List the places where the function in the diagram is not differentiable, and give a brief reason for each.

$x = -2 \rightarrow$ Function is discontinuous at $x = -2$ (Jump)

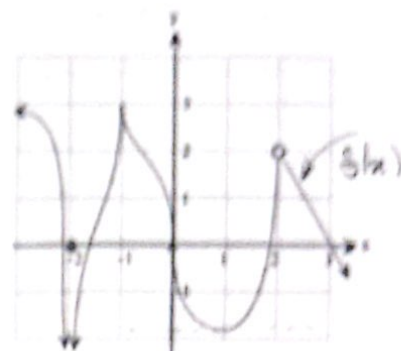
$x = -1 \rightarrow$ Function has a cusp at $x = -1$

$$\lim_{x \rightarrow -1^-} S'(x) = \infty, \lim_{x \rightarrow -1^+} S'(x) = -\infty$$

$x = 0 \rightarrow$ Function has a vertical tangent at $x = 0$

$$\lim_{x \rightarrow 0^-} S'(x) = -\infty, \lim_{x \rightarrow 0^+} S'(x) = \infty$$

$x = 2 \rightarrow$ Function has point discontinuity at $x = 2$



2. Determine the values of each of the following:

a) $f(0)$ Appears to be 0

b) $f(3) = -2$
 $S'(3)$

c) $f(1) = 0$
 $S'(1)$

d) $f(-1) = \text{DNE}$
 $S'(-1)$

3. Differentiate each of the following. Factor and simplify where necessary. Ensure no negative or rational exponents are in final answer.

a) $f(x) = \sqrt[3]{6x - x^3}$

$$S(x) = (6x - x^3)^{\frac{1}{3}}$$

$$S'(x) = \frac{1}{3}(6x - x^3)^{-\frac{2}{3}}(6 - 3x^2)$$

$$S'(x) = \frac{(6 - 3x^2)}{3\sqrt[3]{(6x - x^3)^2}}$$

$$S'(x) = \frac{3(2 - x^2)}{3\sqrt[3]{(6x - x^3)^2}}$$

$$S'(x) = \frac{2 - x^2}{\sqrt[3]{(6x - x^3)^2}}$$

b) $(5x^3 - 3x)^4(5 - 4x)^3$

$$y = (5x^3 - 3x)^4(5 - 4x)^3$$

$$\frac{dy}{dx} = 4(5x^3 - 3x)^3(15x^2 - 3)(5 - 4x)^3 + (5x^3 - 3x)^4 3(5 - 4x)^2(-4)$$

$$= 12(5x^3 - 3x)^3(5x^2 - 1)(5 - 4x)^2 - 12(5x^3 - 3x)^4(5 - 4x)^2$$

$$= [12(5x^3 - 3x)^3(5 - 4x)^2][(5x^2 - 1)(5 - 4x) - (5x^3 - 3x)]$$

$$= [12(5x^3 - 3x)^3(5 - 4x)^2][25x^2 - 20x^2 - 5 + 4x - 5x^3 + 3x]$$

$$\frac{dy}{dx} = [12(5x^3 - 3x)^3(5 - 4x)^2][-25x^3 + 25x^2 + 7x - 5]$$

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4. Analyze the continuity and differentiability of the following function. Show your work and use proper terminology.

$$\textcircled{1} f(x) = \sqrt[5]{2-x}$$

$$= (2-x)^{\frac{1}{5}}$$

$$\lim_{x \rightarrow 2^-} f(x) = 0$$

$$\lim_{x \rightarrow 2^+} f(x) = 0$$

$\therefore f(x)$ is continuous throughout domain

$$f(x) = \sqrt[5]{2-x}$$

$$\textcircled{2} f'(x) = \frac{1}{5}(2-x)^{-\frac{4}{5}}(-1)$$

$$= -\frac{1}{5\sqrt[5]{(2-x)^4}}$$

$$\textcircled{3} \lim_{x \rightarrow 2^-} f'(x) = -\infty$$

$$\lim_{x \rightarrow 2^+} f'(x) = -\infty$$

Therefore, $f(x)$ is non-differentiable because of a vertical tangent at $x=2$. However the function is continuous throughout the domain.

5. Determine the equation of the tangent to the curve $y = \frac{x}{\sqrt{2x+3}}$ at $x=3$.

$$y = x(2x+3)^{-\frac{1}{2}}$$

$$y' = (1)(2x+3)^{-\frac{1}{2}} + x(-\frac{1}{2})(2x+3)^{-\frac{3}{2}}(2)$$

$$y' = \frac{1}{\sqrt{2x+3}} - \frac{2x}{2\sqrt{(2x+3)^3}} \rightarrow = \frac{-2x}{2\sqrt{(2x+3)^3}}$$

$$y = \frac{x}{\sqrt{2x+3}}$$

$$y = 1$$

$$P(3, 1)$$

$$x=3$$

$$y' = \frac{1}{\sqrt{2(3)+3}} + \frac{-2(3)}{2\sqrt{(2(3)+3)^3}}$$

$$y' = \frac{1}{3} - \frac{6}{2(27)}$$

$$y' = \frac{1}{3} - \frac{1}{9}$$

$$y' = \frac{3}{9} - \frac{1}{9}$$

$$y' = \frac{2}{9} = m$$

$$y = \frac{2}{9}x + b$$

$$1 = \frac{6}{9} + b$$

$$\frac{3}{3} - \frac{2}{3} = b$$

$$b = \frac{1}{3}$$

$$y = \frac{2}{9}x + \frac{1}{3}$$

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